CAPSTONE · AI FOR GOOD · WEEK 4

Strategic forecasting and *scenario planning*

From forecasting one future to preparing for four.

Dr. Jan Erik Meidell · Wednesday 4 November 2026 · online · 18:15–20:00 · keep the shared canvas open on a second screen

TONIGHT

One hour forty-five, two exercises

18:18	One number in the chat
18:22	Why forecasts fail: Ørsted 2023, the cone, forecast vs scenario
18:42	The method: Wack, Knight, Schoemaker's ten steps
18:50	Trends, uncertainties, and picking two axes
19:02	Exercise 1 · checking the axes: your team's 2×2 on the screen
19:18	Break · 5 minutes
19:23	Four worlds that work: anchors, tests, signals
19:34	Wind tunnelling: robust, flexible, bet
19:40	Exercise 2 · stress-test Ørsted and Eni, in breakout rooms
19:52	Pitfalls, the report, close

ONE NUMBER IN THE CHAT

EU offshore wind capacity in 2040, in GW

Type one number in the Zoom chat. Do not look it up. Press Enter when I say "now", all at the same time.

The member states' own goal for 2040, agreed by sea basin (updated December 2024): **259–261 GW**.

The point is not who was closest. The point is how far apart your numbers are.

Week 3 asked: what is the best estimate? Week 4 asks: what if the estimate is wrong?

A CAUTIONARY TALE

Ørsted, 2023: excellent forecasts, no scenario

What happened

In autumn 2023 Ørsted cancelled Ocean Wind 1 and 2 off New Jersey and booked DKK 28.4 billion of impairments in the first nine months of 2023 (about €3.8 billion), DKK 19.9 billion of it on Ocean Wind 1. The dividend was suspended for 2023–2025.

Why

Four things moved at once: interest rates (up more than four percentage points in 2022–23), turbine and steel costs, the supply chain, and US permitting and tax-credit timing. Each was visible in 2021. Nobody had put them together in one written, named future and tested the strategy against it.

A forecast asks what is most likely. Ørsted needed to ask what it must be ready for.

THE PATTERN

Three companies, one mistake

COMPANY	THE FORECAST	WHAT HAPPENED
Kodak	Digital cameras stay a niche (its own engineer built the first one in 1975)	Bankruptcy, 2012
Nokia	Smartphones stay a high-end niche above feature phones	Phone business sold to Microsoft, 2013–14
Ørsted	Offshore costs keep falling; the US market opens smoothly	US projects cancelled, 2023

They had

Good data · smart analysts · careful sensitivity analyses · a strong central forecast

They lacked

Several distinct futures · a strategy tested in each · early signals to watch · a plan for being wrong

THE CONE OF UNCERTAINTY

The cone of uncertainty: the further ahead you look, the wider the range of possible futures

What the cone is: draw the possible values of a number (say, Europe's wind capacity) year by year from today. For next year they sit close together; for 2040 they spread far apart. The shape is a cone opening from today, and it tells you where a single forecast still works and where it stops working.

2026–2028

1–3 years · predictable

Projects already financed. A forecast works here.

2030

5 years · emerging

Auctions, rates and supply chains can still move the number a lot.

2040

15 years · divergent

Policy, technology and geopolitics can all change direction. One number is a guess dressed up as an answer.

Your Week 3 learning curve sits in the middle of this cone: central, not certain.

THE DISTINCTION

Forecast vs scenario

	FORECAST	SCENARIO PLANNING
Asks	What is the most likely future?	What futures must we be ready for?
Output	One number, with error bars	Three or four internally consistent stories, each with numbers
Works for	Stable systems, 1–3 years	Systems in transition, 10–30 years
Fails by	Overconfidence: blind to a change of regime	Vague stories that do not change any decision

Your Week 3 projections are forecasts. They are inputs to scenarios, not substitutes for them.

FOUR THINGS GET CALLED "SCENARIOS"

Only one of them is

Forecast

One best estimate of one future. One line, one number.

Sensitivity

One input varied at a time: "what if gas is +20%?" Tests a model; it is not a scenario.

Contingency

A plan for one bad event, the "downside case". Still assumes one future.

Scenario

Several coherent futures built on different axes, used to stress-test a strategy.

Schoemaker (1995) draws this line. Best / base / worst on one axis is a sensitivity analysis, not scenario planning.

CHECK · TYPE F, S, C OR SC IN THE CHAT

Which of the four is it?

1. "EU wind will supply 38% of electricity in 2040." **Forecast**
2. Our LCOE model rerun with capex +15%, everything else unchanged. **Sensitivity**
3. What Eni does if Plenitude's largest auction bid fails. **Contingency**
4. Four 2040 worlds from EU policy ambition × supply-chain openness, each with a story and numbers. **Scenario**
5. Best, base and worst case for the gas price. **Sensitivity, in costume**

WHERE THE METHOD COMES FROM

Shell, 1973, and a distinction from 1921

Pierre Wack at Shell

From the late 1960s Wack's team at Royal Dutch Shell wrote scenarios instead of single forecasts, including one in which oil producers unite and prices jump. In October 1973 the oil embargo came and the crude price roughly quadrupled. Shell had already thought it through. Wack described the method in the *Harvard Business Review* in 1985.

Frank Knight: risk vs uncertainty

Risk can be given a probability, because history is a fair guide: the chance a wind farm produces 5% less than expected in a given year. **Uncertainty** cannot: EU climate policy in 2040. Scenarios are the tool for uncertainty.

Knight, *Risk, Uncertainty and Profit* (1921) · Wack, "Scenarios: Uncharted Waters Ahead", HBR (1985)

YOUR READING · SCHOEMAKER (1995)

Ten steps, and where each one lands

STEP	WHAT YOU DO	WHEN
1–2	Define the scope · identify the stakeholders	Weeks 1–2, done
3–4	Identify the trends · identify the key uncertainties	**Tonight**
5	Construct the scenario themes: the 2×2	**Tonight**
6–7	Check consistency and plausibility · develop the scenarios with numbers	Tonight → Week 5
8–9	Identify research needs · build quantitative models	Phase 2
10	Turn them into decision scenarios	Phase 3

Schoemaker, P.J.H., "Scenario Planning: A Tool for Strategic Thinking", *Sloan Management Review* 36(2), 1995, pp. 25–40.

STEP 3 AND STEP 4

Plan around trends, build scenarios around uncertainties

Trends · fairly predictable

Electrification of transport and heat · turbines keep getting larger · old fossil plants retire on known dates · climate targets already in law · data-centre demand rising

They go into **every** scenario.

Uncertainties · could go either way

Does the EU hold its climate line after 2030? Does the learning rate continue or stall? Do grid and permitting bottlenecks ease? Is the supply chain open or used as a weapon? Do interest rates stay high?

Your two **axes** come from here.

The test: if you are fairly sure which way it goes, it is a trend, not an axis.

THE FILTER

The impact × uncertainty matrix

↑ IMPACT ON THE QUESTION

High impact · low uncertainty

Trends. Put them in every world as a baseline. *Electrification.*

High impact · high uncertainty

Critical uncertainties. Your two axes live here. *EU policy after 2030; the learning rate.*

Low impact · low uncertainty

Ignore. *Office paper consumption.*

Low impact · high uncertainty

Watch, do not build on. *Blade-recycling technology.*

UNCERTAINTY →

STEP 5 · THE HARDEST JUDGEMENT CALL

Picking your two axes

1. **Truly uncertain.** Both ends of the axis must feel plausible in 2040.
2. **High impact on the question.** The answer to "is wind a sound investment?" must differ at each end. Otherwise the axis is decoration.
3. **Independent of each other.** If axis B is partly caused by axis A, you get three worlds, not four.

The orthogonality test: "If axis A lands at one end, does that tell me anything about axis B?" If yes, change one of them.

BEFORE YOU COMMIT

Some forces travel together

	EU POLICY	INTEREST RATES	CARBON PRICE	COST DECLINE
EU policy	—	weak	STRONG	moderate
Interest rates	weak	—	weak	STRONG
Carbon price	STRONG	weak	—	moderate

EU policy and the carbon price cannot be two axes: the carbon price is a policy instrument. High rates raise the cost of capital and slow cost decline, so those two are linked as well.

Make the second axis a **driver**, not an **outcome** of the first.

EXERCISE 1 · 16 MINUTES · THE ORTHOGONALITY COURT

Your team's 2×2 on the screen

1. I open your team's Week 4 block on the canvas. **One person** presents: the two axes, the four world names, the pair you rejected and why. **2 minutes.**
2. The **other team** writes the two **off-diagonal** worlds in the chat, one sentence each (axis 1 high + axis 2 low, and the reverse). **2 minutes.**
3. Verdict, together: did the off-diagonal sentences need contortions? **Pass**, or **swap one axis**, and which driver could replace it. **3 minutes.**

Team 1 first, then Team 2. The verdict goes into your canvas block, in the box "The court's verdict".

VERDICTS

What a pass and a fail look like

Fail · the axes move together

"EU coordination" × "delivery capacity". A fragmented Europe also delivers less, so the off-diagonal worlds need a long story to exist. The second axis is an **outcome** of the first.

Pass · two drivers

"EU policy ambition" × "pace of cost decline". A strong-policy world can still see slow cost decline if supply chains seize; a weak-policy world can still get cheap turbines from global manufacturing.

Drivers that pair well with policy: cost decline · supply-chain geopolitics · interest rates · speed of electrification · grid build-out.

BREAK

Five minutes

Back at 19:23. Cameras on when you return.

A WORKED 2×2 · FROM THE WEEK 4 PAGE

EU policy ambition × pace of cost decline

↑ FAST COST DECLINE

Market-Driven Greening

Weak policy, fast cost decline. Wind wins on price alone; deployment follows corporate power contracts, not auctions.

Renewable Renaissance

Strong policy, fast cost decline. Both engines fire; the limit becomes execution: vessels, ports, people.

Stalled Transition

Weak policy, slow cost decline. Cancellations and delays; gas stays the default. The pure-play is in trouble.

Subsidised Transition

Strong policy, slow cost decline. Wind is built because the state pays for it; margins depend on contract design.

WEAK POLICY ← → STRONG POLICY · ↓ SLOW COST DECLINE

STEP 7 · THE DISCIPLINE OF NUMBERS

Every world needs 3–5 anchors

ANCHOR, 2040	RENAISSANCE	SUBSIDISED	MARKET-DRIVEN	STALLED
Wind share of EU electricity	**55%**	**40%**	**35%**	**22%**
Offshore LCOE (€/MWh)	**42**	**68**	**55**	**85**
EU carbon price (€/t)	**180**	**160**	**60**	**45**

Numbers expose what prose hides. "Fast cost decline" with an LCOE of €70 does not add up.

Illustrative anchors, to show the shape. Today wind is about 17–19% of EU electricity (Ember). Your own anchors need a source and a reason.

STEP 6 · THREE CHECKS

Is each world a scenario, or a mood?

Consistent

Every element fits the axis it sits on. "High carbon price + weak policy" fails: one drives the other.

Distinct

Four different worlds, not one story with four numbers. If all four are "growth", you have missed a break.

Plausible

Each world can unfold from today without a miracle. Start from where Europe is in 2026.

The newspaper test: write the front-page headline of 14 March 2040 in that world. If you cannot, the world is still too vague.

EARLY SIGNALS

How would you know which world is coming?

For each world, two or three signals you could see between 2026 and 2030. Examples for **Renaissance**:

- Offshore auctions clear below €60/MWh in three rounds in a row
- The EU's 2040 climate target is written into law without being weakened
- A large offshore project is commissioned on time and on budget

Igor Ansoff (1975) called these weak signals. In Phase 3 they become the early-warning dashboard for your client (report §17).

FROM SCENARIOS TO STRATEGY

Wind tunnelling

Take a strategy and put it in each world, like a model plane in a wind tunnel. Does it **thrive, survive, struggle or fail**? What breaks first?

Robust

Works in all four worlds. Lower upside, lower downside.

Flexible

Keeps options open: pay a little now for the right to commit later, when the signals are clear.

Bet

Wins big in one world, fails in others. Only with real conviction about that axis.

YOUR TWO COMPANIES

Ørsted and Eni through this lens

Ørsted · pure-play

Sold its oil and gas; everything on offshore wind. **A bet**: wins in Renaissance, exposed in Stalled and when capital is expensive. Little optionality: committed capacity. 2023 is what failing in one world looks like.

Eni · hedged

Keeps oil and gas; grows renewables inside Plenitude, with minority offshore stakes. **Robust and flexible**: rarely first, rarely last. Exposed in Renaissance, where pure-plays take the market.

Which is right depends on which worlds you take seriously. That is the question your report answers.

EXERCISE 2 · 8 MINUTES · BREAKOUT ROOMS

Stress-test both companies in your four worlds

1. Use **your team's** four worlds, after tonight's verdict.
2. In the canvas table **"The stress test"**: one row per world, Ørsted and Eni each scored **thrive · survive · struggle · fail**.
3. One reason per cell, naming a real feature of the company: "pure-play offshore", "oil and gas cash flow", "minority stakes".
4. Last line: in which world does the answer **flip**?

Back in the main room at 19:48. One person per team reads the flip line aloud.

READ-OUT

Where does the answer flip?

If Ørsted wins in one world and Eni in the other three, that is not yet a recommendation. It depends on the client: a long-horizon pension fund and a growth investor read the same table differently.

The table you just made is the first draft of report §15. In Week 9 you score it again with your Phase 2 numbers.

FIVE PITFALLS

What to avoid in Phase 1

1. **Best / base / worst.** That is sensitivity analysis on one axis.
2. **Axes that move together.** Policy × carbon price gives two worlds, not four.
3. **The official future in disguise.** One world equals the EU target; the other three exist to flatter it.
4. **No numbers.** Without anchors a scenario is a mood.
5. **No strategy test.** Scenarios that never meet Ørsted and Eni are just stories.

THIS WEEK

From tonight to the report

TONIGHT PRODUCED	IT GOES INTO
The scored force list · the trends	Report §3 · the structural forces
Your two axes, the verdict, four named worlds with anchors	Report §6 · two critical uncertainties and the four worlds
The stress test	A first draft of §15 (Phase 3)

On the Week 4 page: Exercises 1–3, self-paced, before next Wednesday. Phase 1 (§1–§7) is due **Sunday 15 November, 23:59**.

Read Schoemaker if you have not: the ten steps (pp. 27–34) and the pitfalls (p. 39).

CLOSE

Commit to a method, *not to a number*

Good consultants are not the ones who predict 2040 correctly. They are the ones whose recommendation still holds when 2040 turns out differently.

Questions on the material: the forum of the week. Anything personal: the message panel, bottom right of every page.